LINEAR ALGEBRAS WITH ASSOCIATIVITY NOT ASSUMED

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1. The complete structure of linear associative algebras was known to depend upon the division algebras. When the reference field F is an algebraic field, H. Hasse has recently proved that every normal[1] division algebra is cyclic. This perfection of the theory of associative algebras justifies attention to non-associative algebras.

Known examples of non-associative division algebras are Cayley's algebra of order 8, and the writer's[2] commutative algebras of orders 3 and $2n$ (§15). Many new division algebras of order 4 are given here by Theorems 2 and 3.

In §§7–11 we determine all types of algebras of order 3 having a principal unit (or modulus) denoted by 1. Except for special values of the parameters, these algebras are simple. It is known that every associative simple algebra of order 3 is a division algebra.

Thus the structure theorems for associative algebras fail in general for non-associative algebras. Similarly for other properties. Consider the algebra A of order 4 with $e_1^2 = e_2$, $e_2e_1 = e_3$, and all further $e_ie_j = 0$. For $X = x_0 + \Sigma x_ie_i$, evidently $(X - x_0)(X - x_0)^2 = 0$, so that A has the left rank 3. But its right rank is 4 since $1, e_1, e_1^2 = e_2, e_1^2\, e_1 = e_3$ are linearly independent. Algebra (37) with 13 parameters also has left and right ranks 3 and 4.

Part I. Rank 2

2. In case the field F has a modulus p, assume that $p \neq 2$.

Lemma 1. *If* $1, u, v$ *are linearly independent with respect to* F, *and* $u^2 = a + cu$, $v^2 = b + dv$, *then*

$$uv + vu = du + cv + f, \qquad f \text{ in } F.$$

Write t for $a + b + cu + dv$. Then

$$(u + v)^2 = t + uv + vu = r + s(u + v),$$

$$(u - v)^2 = t - uv - vu = R + S(u - v).$$

Addition yields $s + S = 2c$, $s - S = 2d$. Subtraction gives Lemma 1.

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[1] If only the numbers of F are commutative with every element.

[2] *Linear Algebras*, Cambridge Tract No. 16, p. 14, p. 69 (p. 17 for the characteristic equations); *On triple algebras and ternary cubic forms*, Bull. Amer. Math. Soc., vol. 14 (1907–8), pp. 160–169, p. 169; *Linear algebras in which division is always uniquely possible*, Trans. Amer. Math. Soc., vol. 7 (1906), pp. 370–390; *On commutative linear algebras in which division is always uniquely possible*, ibid., pp. 514–522.

3. After adding scalars (numbers of F) to the initial units, we obtain a basis $1, e_1, \cdots, e_n$, where $e_i^2 = a_i$. By Lemma 1,

$$e_ie_j + e_je_i = b_{ij} \qquad (i \neq j).$$

We get e_je_i from the latter and

$$e_ie_j = c_{ij} + \sum_{k=1}^{n} c_{ijk}e_k \qquad (i < j).$$

Using summations for $i = 1, \cdots, n$, let

$$X = \xi + \Sigma x_ie_i, \qquad Y = \eta + \Sigma y_ie_i, \qquad XY = P_0 + \Sigma P_ie_i.$$

Let Σ' denote summation for $i, j = 1, \cdots, n$ with $i < j$. Then

$$P_0 = \xi\eta + \Sigma x_iy_ia_i + \Sigma' b_{ij}x_jy_i + \Sigma'(x_iy_j - x_jy_i)c_{ij},$$

$$P_k = \xi y_k + \eta x_k + \Sigma'(x_iy_j - x_jy_i)c_{ijk} \qquad (k = 1, \cdots, n).$$

The right determinant $\Delta(X)$ is defined to be the determinant of the coefficients of $\eta, y_1, \cdots, y_n$ in $P_0, P_1, \cdots, P_n$. Multiply the first column by $-\xi$ and the $(j+1)$-st column by x_j (for $j = 1, \cdots, n$) and add. The sum is the value of P_k for $\eta = -\xi$, $y_j = x_j$. For $k \geqq 1$, P_k evidently vanishes. Since Y becomes $-X'$, where X' is the conjugate of X, P_0 becomes the negative of the norm of X.

THEOREM 1. *The right determinant $\Delta(X)$ and left determinant $\Delta'(X)$ of any algebra of rank 2 are both divisible by the norm $N(X)$.*

The quotient is obtained at once from the cofactor of any element in the first row (cf. §5).

4. **Division algebras of rank 2.** We assume that if any product of two factors is zero, then one factor is zero.

LEMMA 2. *If $1, u, v$ are linearly independent in F, then $1, u, v, uv$ are linearly independent in F.*

We can choose m and n in F such that

$$U = u - m, \qquad V = v - m, \qquad U^2 = a, \qquad V^2 = k.$$

By Lemma 1, $UV + VU = 2c$. Take $e_1 = U$, $e_2 = V - ca^{-1}U$. Then

$$e_1^2 = a, \qquad e_2^2 = b, \qquad e_1e_2 + e_2e_1 = 0.$$

If $a = k^2$, k in F, then $(e_1 - k)(e_1 + k) = 0$, $e_1 = \pm k$, whereas 1 and e_1 are linearly independent in F. Hence

(1) $\qquad a \neq$ square in F, $\qquad b \neq$ square in F.

If $1, u, v, uv$ are linearly dependent in F, the same is true of $1, U, V, UV$ and of $1, e_1, e_2, e_1e_2$. Hence $e_1e_2 = r + se_1 + te_2$, where r, s, t are in F. Then

$$(e_1 - t)\left(e_2 + \frac{r + st}{t^2 - a}e_1 + \frac{rt + as}{t^2 - a}\right) = 0,$$

whereas neither factor is zero. This proves Lemma 2.

Since we may take e_1e_2 as the fourth basal element e_3,

$$(2) \qquad e_1^2 = a, \qquad e_2^2 = b, \qquad e_1e_2 = e_3, \qquad e_2e_1 = -e_3.$$

Write $e_i^2 = c_i + 2m_ie_i$ for $i \geqq 3$. Apply Lemma 1 with $(u, v) = (e_1, e_i)$, (e_2, e_i), (e_i, e_j) in turn. We get

$$(3) \qquad \begin{gathered} e_1e_i + e_ie_1 = g_i + 2m_ie_1, \qquad e_2e_i + e_ie_2 = h_i + 2m_ie_2 \qquad (i \geqq 3), \\ e_ie_j + e_je_i = t_{ij} + 2m_ie_j + 2m_je_i \qquad (i, j \geqq 3, i \neq j). \end{gathered}$$

Write $x = \xi + \Sigma x_ie_i$. Squaring $x - \xi$, we get

$$x^2 = 2x\left(\xi + \sum_{j=3}^{n} x_jm_j\right) - N,$$

$$N = \xi^2 + 2\xi\Sigma x_jm_j - ax_1^2 - bx_2^2 - \Sigma c_ix_i^2 - \Sigma g_jx_1x_j - \Sigma h_jx_2x_j - \Sigma' t_{ij}x_ix_j,$$

where the summations are for $i \geqq 3, j \geqq 3$, while $i < j$ in Σ'. Thus (2) and (3) imply that every element x of the algebra satisfies a quadratic equation. Evidently N is the norm xx' of x. Also,

$$(4) \qquad x^2 \text{ is scalar if and only if } \xi = -\sum_{j=3}^{n} x_jm_j.$$

We may choose a basis satisfying (2) and $e_3^2 =$ scalar. Proof is needed only when $m_3 \neq 0$. Use the basis

$$1, \qquad e_1, \qquad E_2 = x = \xi + \Sigma x_ie_i, \qquad E_3 = e_1E_2, \qquad \cdots,$$

where (4) holds. Then

$$e_1E_2 + e_2E_1 = 2ax_1 + \sum_{i=3}^{n} g_ix_i$$

becomes zero by choice of x_1. Let

$$e_1e_i = k_i + \sum_{j=1}^{n} c_{ij}e_j \qquad (i \geqq 3).$$

Then

$$E_3 = s + \xi e_1 + x_2e_3 + \sum_{j=1}^{n}\left(\sum_{i=3}^{n} x_ic_{ij}\right)e_j, \quad s = ax_1 + \sum_{i=3}^{n} x_ik_i.$$

By (4), E_3^2 is scalar if

$$s + m_3(x_2 + \sum_{i=3}^{n} x_ic_{i3}) + \sum_{j=4}^{n} m_j\left(\sum_{i=3}^{n} x_ic_{ij}\right) = 0.$$

This and the former condition hold if, for example,

$$x_i = 0\ (i \geqq 4), \qquad x_3 = 2, \qquad x_1 = -g_3/a, \qquad m_3x_2 = g_3 - 2k_3 - 2\sum_{j=3}^{n} m_jc_{3j}.$$

LEMMA 3. *There exists a basis satisfying* (2) *and* (3) *with* $m_3 = 0$.

5. **Case** $n = 3$. Employ (2) and

$$(5)\qquad \begin{aligned} e_1e_3 &= c_0 + c_1e_1 + c_2e_2 + c_3e_3, \qquad & e_2e_3 &= d_0 + d_1e_2 + d_2e_2 + d_3e_3,\\ e_3^2 &= f, & e_3e_1 &= g - e_1e_3, & e_3e_2 &= h - e_2e_3, \end{aligned}$$

which follow from (3). As in Section 3, $\Delta(x)$ is

$$\begin{vmatrix} \xi & ax_1 + (g - c_0)x_3 & bx_2 + (h - d_0)x_3 & c_0x_1 + d_0x_2 + fx_3 \\ x_1 & \xi - c_1x_3 & -d_1x_3 & c_1x_1 + d_1x_2 \\ x_2 & -c_2x_3 & \xi - d_2x_3 & c_2x_1 + d_2x_2 \\ x_3 & -x_2 - c_3x_3 & x_1 - d_3x_3 & c_3x_1 + d_3x_2 + \xi \end{vmatrix},$$

which is the product of

$$(6)\qquad -N(x) = -\xi^2 + ax_1^2 + bx_2^2 + fx_3^2 + gx_1x_3 + hx_2x_3$$

and the quotient[3] by $-x_3$ of the minor obtained by deleting the first row and last column. This quotient is

$$(7)\qquad \begin{aligned} &-\xi^2 - c_3\xi x_1 - d_3\xi x_2 + (c_1 + d_2)\xi x_3 + c_2x_1^2 + (d_2 - c_1)x_1x_2 \\ &\quad -d_1x_2^2 + (c_3d_2 - c_2d_3)x_1x_3 + (c_1d_3 - c_3d_1)x_2x_3 + (c_2d_1 - d_1d_2)x_3^2. \end{aligned}$$

Similarly, the left determinant $\Delta'(x)$ is the product of (6) by the function derived from (7) by changing the sign of ξ.

THEOREM 2. *The algebra* R *over* F *defined by* (2) *and* (5) *is a division algebra if and only if neither* (6) *nor* (7) *is zero for values of* ξ, x_1, x_2, x_3 *in the field not all zero. The forms* (6) *and* (7) *are identical if and only if* R *is the algebra* S *defined by* (2) *and*

$$(8)\qquad \begin{aligned} e_3^2 &= -ab, & e_1e_3 &= c_0 + ae_2, & e_3e_1 &= -e_1e_3, \\ & & e_2e_3 &= d_0 - be_1, & e_3e_2 &= -e_2e_3. \end{aligned}$$

In R *the product of the norms of any two elements is the norm of their product if and only if* R *is the algebra* Q *defined by* (2) *and* (8) *with* $c_0 = d_0 = 0$, *whence* Q *is the generalized quaternion algebra and obeys the associative law.*

In the products of (6) and (7) by -1 and -4, respectively, we complete the square on ξ, then on x_1, and finally on x_2. We obtain forms having only square terms with the coefficients

$$(6')\qquad 1, -a, -b, -f + g^2/(4a) + h^2/(4b)\ ;$$

$$(7')\qquad 1, -A, -B, D + C^2/B,$$

[3] Or the quotient by x_2 of the minor obtained by deleting the first row and third column, etc.

where

$$A = c_3^2 + 4c_2, \qquad B = d_3^2 - 4d_1 - U^2/A, \qquad U = 2(c_1 - d_2) - c_3d_3,$$

$$W = c_1c_3 + 2c_2d_3 - c_3d_2, \qquad C = 2c_3d_1 - c_1d_3 + d_2d_3 + UW/A,$$

$$D = 4(c_1d_2 - c_2d_1) - (c_1 + d_2)^2 + W^2/A.$$

When F is the field of all real numbers, the condition for a division algebra is that the numbers (6′) and (7′) be all positive.

When F is any field, $x^2 - ay^2 - bz^2 + mw^2$ is not zero for values of $x, \cdots, w$ in F not all zero if and only if a is not a square in F, b is not represented by $x^2 - ay^2$, nor $-m$ by $x^2 - ay^2 - bz^2$, for values in F of the variables. The final condition is redundant[4] if $m = ab$.

The case $c_3 = d_3 = c_1 = d_2 = g = h = 0, f = -ab$, yields the

COROLLARY. *The algebra defined by* (2) *and*

$$e_1e_3 = c_0 + ce_2, \quad e_2e_3 = d_0 - de_1, \quad e_3^2 = -ab, \quad e_3e_1 = -e_1e_3, \quad e_3e_2 = -e_2e_3$$

is a division algebra if and only if a and c are not squares in F, b is not represented by $x^2 - ay^2$, *and d is not represented by* $x^2 - cy^2$. *When* $c = a$, $d = b$, *this algebra is S of Theorem* 2.

6. **Normalization.** Employ a new basis of R:

(9) $$1, \quad E_1, \quad E_2, \quad E_3 = E_1E_2, \quad E_1^2 \text{ and } E_2^2 \text{ scalar.}$$

By (4)

(10) $$E_1 = \Sigma x_ie_i, \qquad E_2 = \Sigma y_ie_i \qquad (i = 1, 2, 3).$$

By Lemma 2, (9) is a basis if and only if the x_i are not proportional to the y_i. The condition for $E_1E_2 + E_2E_1 = 0$ is $C(x_i, y_i) = 0$, viz.,

(11) $$2t + g(x_1y_3 + x_3y_1) + h(x_2y_3 + x_3y_2) = 0, \quad t = ax_1y_1 + bx_2y_2 + fx_3y_3.$$

We get $E_3 = f_0 + \Sigma f_ie_i$, where

$$f_0 = t + c_0x_1y_3 + (g - c_0)x_3y_1 + d_0x_2y_3 + (h - d_0)x_3y_2,$$

(12) $$f_i = c_iL + d_iM \quad (i = 1, 2), \qquad f_3 = c_3L + d_3M + N,$$

$$L = x_1y_3 - x_3y_1, \qquad M = x_2y_3 - x_3y_2, \qquad N = x_1y_2 - x_2y_1.$$

We require that E_3^2 be scalar, viz., $f_0 = 0$. Next $C(x_i, f_i) = 0$ is the condition for $E_1E_3 + E_3E_1 = 0$. We now have three linear homogeneous equations in y_1, y_2, y_3. The determinant D of their coefficients is a quartic form in the x_i. Assume that $D = 0$ has solutions not all zero in F. Then our three equations hold if the y_i are the cofactors of the elements of the third row of D. For these values (quadratic functions of the x_i) of the y_i, the condition $C(y_i, f_i) = 0$

[4] L. E. Dickson, *Algebren und ihre Zahlentheorie*, Zürich, 1927, p. 47.

for $E_2E_3 + E_3E_2 = 0$ becomes an equation in the x_i of degree 5. This has solutions in common with $D = 0$, but not necessarily in F. Hence in general we may normalize algebra R so that $g = h = 0$.

Consider the normalization of algebra S defined by (2) and (8). Here (11) and $f_0 = 0$ become

$$(13) \qquad ax_1y_1 + bx_2y_2 - abx_3y_3 = 0, \qquad c_0L + d_0M = 0.$$

When (13) hold, the new basis (9) satisfies equations (2) and (8) written in capital letters if and only if

$$(14) \qquad A = ax_1^2 + bx_2^2 - abx_3^2, \qquad B = ay_1^2 + by_2^2 - aby_3^2,$$

$$(15) \qquad AB + ab(aL^2 + bM^2 - N^2) = 0,$$

$$(16) \qquad C_0 = ab(x_2L - x_1M - x_3N) + c_0(x_1N + bx_3M) + d_0(x_2N - ax_3L),$$

while D_0 is derived from (16) by replacing x_i by y_i. Using the definitions (12) and (14), we find that the left member of (15) reduces to the square of the first sum in (13). Hence we may discard (15).

To make $C_0 = 0$ for solutions of (13), we have three linear homogeneous equations in the y_i; the determinant of their coefficients is a quartic form in the x_i, which must vanish.

For the special case $c_0 = d_0 = 0$, C_0 and D_0 are identically zero by (12). Hence *every normalization of a generalized quaternion algebra is made by the new units* (10) *and* $E_3 = E_1E_2$, *where the* x_i *are not proportional to the* y_i *and satisfy the single equation* (13_1); *the new parameters are* (14).

7. Triple algebras of rank 2. By the first two sentences of §3,

$$e_1^2 = a, \qquad e_2^2 = b, \qquad e_1e_2 = c + de_1 + fe_2, \qquad e_2e_1 = g - de_1 - fe_2.$$

If $d = f = 0$, we have (17) with $k = 0$. If d and f are not both zero, we may interchange the e's, if necessary, and take $d \neq 0$. Write

$$E_1 = de_1 + fe_2, \qquad E_2 = d^{-1}e_2,$$

$$E_1E_2 = c + bf/d + E_1, \qquad E_2E_1 = g + bf/d - E_1.$$

Hence every triple algebra of rank 2 is equivalent to[5]

$$(17) \qquad e_1^2 = a, \quad e_2^2 = b, \quad e_1e_2 = c + ke_1, \quad e_2e_1 = g - ke_1 \qquad (k = 0 \text{ or } 1).$$

Every transformation of units for which the E_i^2 are scalar is

$$(18) \qquad E_1 = re_1 + se_2, \qquad E_2 = te_1 + we_2, \qquad D = rw - st \neq 0.$$

Then $E_1E_2 = kDe_1 +$ scalar. Let (17) hold in capital letters. Then $kDe_1 = KE_1$, whence $kD = Kr$, $Ks = 0$. Hence an algebra (17) with $K = 0$ is not equivalent to one with $k \neq 0$.

[5] If $k = 0$, we may take $c = 0$ unless $a = b = c + g = 0$.

I. $k = 1$. Then $K = 1$, $s = 0$, $w = 1$, $r \neq 0$. Then $E_2E_1 = G - E_1$, $G = r(g + ta)$, $E_1E_2 = C + E_1$, $C = r(c + ta)$.

I$_1$. $a \neq 0$. We make $G = 0$ by choice of t. Hence we may take $g = 0$. Then $G = 0$ requires $t = 0$, $E_1 = re_1$, $E_2 = e_2$. Then

$$E_1^2 = r^2a, \qquad E_2^2 = b, \qquad E_1E_2 = rc + E_1, \qquad E_2E_1 = -E_1.$$

If $c \neq 0$, take $rc = 1$. Thus

$$(19) \qquad e_1^2 = a, \qquad e_2^2 = b, \qquad e_1e_2 = c + e_1, \qquad e_2e_1 = -e_1 \qquad (c = 0 \text{ or } 1),$$

with no further normalization if $c = 1$. But if $c = 0$, we may remove any square factor from a.

I$_2$. $a = 0$. Then $G = rg$, $C = rc$, $A = r^2a$, $B = b + t(c + g)$. If $c + g \neq 0$, we may take $b = 0$ and a non-vanishing one of c and g equal to 1. If $c + g = 0$, $c \neq 0$, we may take $c = 1$. If $c = g = 0$, we can remove a square factor from a. No further normalization is possible.

For any e in algebra (19), $(1, e)$ is a sub-algebra which is not invariant. Every sub-algebra of order 2 lacking 1 has a basis in which the pairs of coefficients of e_1 and e_2 are not proportional and hence a basis $r + e_1$, $s + e_2$, with

$$c = 2r, \qquad r(s - 1) = 0, \qquad r^2 = a, \qquad s^2 = b.$$

Either $c = a = r = 0$, $b = s^2$, or $c = b = s = 1$, $r = 1/2$, $a = 1/4$. It is invariant by

LEMMA 4. *If S is a sub-algebra of order n of an algebra A of order $n + 1$ and if A has a modulus 1, but 1 is not in S, then S is invariant in A.*

For, $A = (1, S)$, $S^2 \leqq S$. Hence $AS \leqq S$, $SA \leqq S$.

If $c = a = 0$, (19) has the single invariant sub-algebra $[e_1]$ of order 1, otherwise none.

A simple algebra is one having no proper invariant sub-algebra. *Hence* (19) *is simple unless $c = a = 0$ or $c = b = 1$, $a = 1/4$.* Algebra (19) is associative if and only if $a = c = 0$, $b = 1$.

Part II. Triple algebras of rank 3

8. Employ the new basis

$$(20) \qquad 1, \quad E_1 = a + be_1 + ce_2, \quad E_2 = e + fe_1 + ge_2, \quad D = bg - cf \neq 0.$$

The commutative case shows the existence of a relation

$$(21) \qquad E_2E_1 - E_1E_2 = D(e_2e_1 - e_1e_2).$$

We may choose e_1 so that 1, e_1, e_1^2 are linearly independent in F. Hence we may take

$$(22) \qquad \begin{aligned} e_1^2 &= e_2, \qquad e_1e_2 = h + je_1 + ke_2, \\ e_2^2 &= m + ne_1 + pe_2, \qquad e_2e_1 = r + se_1 + te_2. \end{aligned}$$

For any new basis (20) we will have $E_1^2 = E_2$ if

$$e = a^2 + bc(h + r) + c^2m,$$

$$f = 2ab + bc(j + s) + c^2n, \tag{23}$$

$$g = 2ac + b^2 + bc(k + t) + c^2p.$$

Let (22′) denote the multiplication table for E_1, E_2; it is of the form (22) with each letter replaced by the corresponding capital letter. Using also (21), we get

$$R - H + (S - J)E_1 + (T - K)E_2 = D[r - h + (s - j)e_1 + (t - k)e_2]. \tag{24}$$

Solving (20), we get

$$De_1 = ce - ag + gE_1 - cE_2, \qquad De_2 = af - be - fE_1 + bE_2. \tag{25}$$

Elimination of e_1 and e_2 between (24) and (25) gives

$$S - J = g(s - j) - f(t - k), \qquad T - K = -c(s - j) + b(t - k). \tag{26}$$

We can always[6] make $S = J$. Proof is needed only when $s \neq j$. By (26), $S = J$ if

$$g = fu, \qquad u = (t - k)/(s - j).$$

By (23), this holds if $2a(c - bu)$ equals a known quadratic form in b, c. This condition is satisfied by choice of a in terms of b and c if $c - bu \neq 0$. Then $D = -f(c - bu) \neq 0$ if $f \neq 0$; we must therefore avoid the values for which f and g in (23) are both zero.

Hence we may take $s = j$. This is not altered if we replace e_1 by $e_1 - k/3$, which yields $K = 0$. Hence we may take

$$s = j, \qquad k = 0. \tag{27}$$

Apply transformation (20) subject to (23). By (24), $S = J$ if $ft = 0$. There are two cases.

9. Let $t \neq 0$. Then $f = 0$, $D = bg \neq 0$. By (23_2),

$$-2a = 2cj + c^2n/b. \tag{28}$$

Then

$$S = J = e + gj + gcn/b, \qquad K = a + c(p - j) - c^2n/b.$$

Eliminating a by (28), we see that $K = 0$ only when

$$c = 0 \qquad \text{or} \qquad 2p - 4j - 3cn/b = 0. \tag{29}$$

[6] But we can make $T = K$ only when a certain invariant is not zero, C. C. MacDuffee, *Invariantive characterizations of linear algebras with the associative law not assumed*, Trans. Amer. Math. Soc., vol. 23 (1922), pp. 135–150, p. 147. In his equation below (27), the final product should be $(b_2 - c_1 - d_1)(c_1 - d_1)(c_2 - d_2)^2 - b_1(c_2 - d_2)^3$, while α_1^3 should be α_2^3.

Now

$$bT = ab + bc(p - j) + b^2t - c^2n$$

reduces by (28) and (29) to

$$T = bt. \tag{30}$$

By choice of b, we may take $T = 1$. To preserve this normalization $t = 1$, we must take $b = 1$. Since

$$N = g^2n/b, \qquad P - 2J = -g(p - 2j), \tag{31}$$

no algebra of one of the following four types is equivalent to one of a different type.

I or II. $n \neq 0$, $p = 2j$, or $n = 0$, $p \neq 2j$. Then $c = 0$ by (29), whence $a = 0$ by (28) and our transformation (20) is the identity.

III. $n \neq 0$, $p \neq 2j$. Then (29) and (28) uniquely determine $c \neq 0$ and a, whence our transformation yields a unique new algebra.

IV. $n = 0$, $p = 2j$. Our transformation involves a single parameter c which is available for further normalization.

10. Let $t = 0$. Then $S = J$, $T = K$ by (26) and (27). After divisions by D, we find that

$$K = 3a + (p + j)c, \qquad J = -3a^2 - 2ac(p + j) + b^2j + bc(n + h + r) + c^2(m - pj),$$

$$P = 6a^2 + 4ac(p + j) + b^2p + bc(2h + 2r - n) + c^2(p^2 + 2pj + 2m).$$

To preserve the normalization $k = 0$, take $K = 0$ and eliminate a. Then J and P become quadratic forms in b, c, while

$$27N = 27nb^3 + 18w^2b^2c - 27wnbc^2 + (2w^3 + 27n^2)c^3,$$

where $w = p - 2j$. The expressions for H, M, R are more complicated, although $R - H = D(r - h)$ by (24). If F is the field of all real numbers, we may take $N = 0$. For a different F, normalization depends on representation by binary forms.

11. Let A be the algebra defined by (22) and (27). Any sub-algebra of order 2 without the modulus 1 is $(u + e_1, v + e_2)$, where

$$v = -u^2, \qquad h = u(v + j), \qquad r - h = tv, \qquad m = nu + v^2 + pv.$$

It is invariant in A by Lemma 4. Elimination of u and v yields two conditions on the constants of A.

Let $W = x + ye_1 + ze_2$ generate an invariant sub-algebra of order 1 of A. We may take $y = 1$ or $y = 0$ and $z = 1$. In the latter case, elimination of $x = -j$ shows that the conditions on A are

$$h = n = 0, \qquad r = -jt, \qquad m = j(j - p).$$

If $y = 1$, the conditions on A are $r = h$, $t = 0$ and the results of eliminating x and z between

$$x = z^2h, \quad z(x + zj) = 1, \quad x(j + zn) = h + zm, \quad z(j + zn) = x + zp.$$

If none of these special sets of conditions holds, A is simple.

Part III. Algebras of order 4 and left rank 3

12. Let xx^2 be a linear combination of $1, x, x^2$ for every element x. Let $1, e, e^2$ be linearly independent. By choice of r in the field F, $e_1 = e + r$ satisfies the first two equations in (33). Let $e_1e_3 = a_0 + \Sigma a_ie_i$. For

$$E = -a_1 - a_2a_3 - a_2e_1 + e_3, \tag{32}$$

e_1E is linear in E. Take E as a new e_3. Hence

$$\begin{aligned} &e_1^2 = e_2, && e_1e_2 = a + be_1, && e_1e_3 = a_0 + a_3e_3, && e_2e_1 = \Sigma b_ie_i, \\ &e_2^2 = \Sigma c_ie_i, && e_2e_3 = \Sigma d_ie_i, && e_3e_1 = \Sigma f_ie_i, \\ & && e_3e_2 = \Sigma g_ie_i, && e_3^2 = \Sigma h_ie_i, \end{aligned} \tag{33}$$

where $e_0 = 1$. Write

$$\begin{aligned} X = se_1 + te_2 + we_3, \qquad & X^2 = K + Le_1 + Me_2 + Pe_3, \\ XX^2 = \text{scalar} + Re_1 + Te_2 + Ve_3. & \end{aligned} \tag{34}$$

The algebra will have left rank 3 if 1 and these three elements are linearly dependent. This is true if and only if

$$\begin{vmatrix} s & L & R \\ t & M & T \\ w & P & V \end{vmatrix} = 0. \tag{35}$$

Since M has the term s^2, while L and P are linear in s, the minors of V and R are not zero identically. We here assume that the minor of T is zero, whence $L \equiv sq$, $P \equiv wq$. Then (35) is the product of $M - tq \not\equiv 0$ by $sV - wR \equiv 0$. By using the conditions which follow readily from $L \equiv sq$, $P \equiv wq$, and from the terms free of t in $sV - wR$, we find that the latter is the product of q by a determinant which is evidently zero. Hence the left rank is 3 for the algebra

$$\begin{aligned} &e_1^2 = e_2, \quad e_1e_2 = a + be_1, \quad e_2e_1 = b_0 + b_1e_1 + b_2e_2, \quad e_2^2 = c_0 + c_2e_2, \\ &e_2e_3 = d_0 + d_2e_2 + b_1e_3, \quad e_1e_3 = a_0 + a_3e_3, \quad e_3e_2 = g_0 + g_2e_2 + be_3, \\ &e_3e_1 = f_0 + f_1e_1 + f_2e_2 - a_3e_3, \quad e_3^2 = h_0 + h_2e_2 + f_1e_3. \end{aligned} \tag{36}$$

Then X^2, e_2X and Xe_2 are linear functions of $1, e_2, X$. Hence if Y is any element, $1, e_2, Y$ give a basis of a sub-algebra. Hence algebra (36) has left rank 3 and right rank 3. For, from the linear function of $1, e_2, Y$ giving Y^2Y (or YY^2) we may eliminate e_2 by means of $Y^2 = g + he_2 + jY$, $h \neq 0$.

13. Conversely, for every element Y of an algebra A of order 4 over a field F, let 1, E, Y give a basis of a sub-algebra, where E is fixed and is not a multiple of 1. The right and left ranks of A are $\leqq 3$. Exclude A's of rank 2. We choose an element e not satisfying a quadratic equation in A. The case $Y = E$ shows that E satisfies a quadratic equation in A. Hence 1, e, E are linearly independent in F. The case $Y = e$ gives $e^2 = r + se + tE$, $t \neq 0$. Write

$$e_1 = e - s/2, \qquad e_2 = e_1^2 = r + s^2/4 + tE.$$

Then 1, e_2, Y give a basis of our sub-algebra. Take (32) as a new e_3. We find that 1, e_2, Y form an algebra for every Y in A if and only if A is identical with algebra (36) with the second equation replaced by $e_1e_2 = a + be_1 + ce_2$.

14. Modify §12 by taking $L \equiv sQ$, $P \equiv st + wQ$. Then (35) holds (whence the left rank is 3) if and only if algebra (33) becomes

$$e_1^2 = e_2, \quad e_1e_2 = a + be_1, \quad e_1e_3 = a_0, \quad e_2e_1 = b_0 + b_1e_1 + b_2e_2 + e_3,$$

$$(37) \quad e_2^2 = c_0 + (b + b_1)e_2, \quad e_2e_3 = d_0 + b_1e_3, \quad e_3^2 = h_0 + f_1e_3,$$

$$e_3e_2 = g_0 + f_1e_2 + be_3, \quad e_3e_1 = f_0 + f_1e_1 + f_2e_2.$$

The right rank is 4, as shown by the right minimum equation for e_1. Note the sub-algebra (1, e_2, e_3).

Part IV. Division algebras

15. Let J be a root of a cyclic equation

$$x^n - c_1x^{n-1} + \cdots + (-1)^nc_n = 0,$$

i.e., an equation irreducible in a field F having the roots

$$J, J' = \theta(J), J'' = \theta(J') = \theta^2(J), \cdots, J^{(n-1)} = \theta^{n-1}(J), [\theta^n(J) = J],$$

where superscripts on θ denote iteratives (and not powers), while θ is a polynomial with coefficients in F. Write $B' = B(J)$, $B'' = B(J')$, etc. Define an algebra with the elements $A + BI$, where A and B are polynomials in J with coefficients in F, with the law of multiplication

$$(38) \quad (A + BI)(X + YI) = R + SI, \; R = AX + gBY'J, \; S = BX' + AY,$$

where g is a number $\neq 0$ of F. Both distributive laws hold.

Given R, S and X, Y not both zero, we can find A, B uniquely if

$$(39) \qquad \begin{vmatrix} X & gY'J \\ Y & X' \end{vmatrix} \neq 0.$$

This holds if $Y = 0$. Next, let $Y \neq 0$, $XX' = gYY'J$. This fails if $n = 2$

since J is not in F. If $n \geqq 3$, we take the product of its conjugates, write $N(X)$ for the norm $XX' \cdots X^{(n-1)}$ of X, and get

$$N^2(X) = g^n c_n N^2(Y),$$

which fails if $g^n c_n$ is not the square of the norm of a number of $F(J)$.

Given R, S and A, B not both zero, we seek X, Y. But

$$C \equiv gBS'J - A'R = gBB'JX'' - AA'X. \tag{40}$$

If $n = 2$, then $X'' = X$ and we obtain X uniquely and then Y. If $n \geqq 3$, let D denote the determinant of the coefficients of $X, X', \cdots, X^{(n-1)}$ in $C, C', \cdots, C^{(n-1)}$.

If n is odd, we find that

$$D = -N^2(A) + g^n c_n N^2(B). \tag{41}$$

But if n is even, $n \geqq 4$, we find that $D = UU'$, where

$$U = -N(A) + g^{n/2} N(B) JJ'' \cdots J^{(n-2)}. \tag{42}$$

Transpose and multiply by the conjugate equation.

THEOREM 3. *Let $(-1)^n c_n$ be the constant term of a cyclic equation of degree n over a field F. The algebra in which multiplication is defined by* (38) *is a division algebra if $n = 2$, and when $n \geqq 3$ if $g^n c_n$ is not the square of the norm of a number of the field $F(J)$.*

For $n = 2$, the algebra over F has the basis $1, J, I, K$, where

$$\begin{aligned} &J^2 = cJ - d, \quad I^2 = gJ, \quad JI = K, \quad K^2 = dgJ, \quad IK = gd, \quad KJ = dI, \\ &IJ = cI - K, \quad JK = cK - dI, \quad KI = gcJ - gd. \end{aligned} \tag{43}$$

This is a division algebra if $x^2 - cx + d = 0$ is irreducible in F; it is remarkable that there is no further condition.

A generalization of (38) is

$$\begin{aligned} R &= AX + (aBY + bBY' + cB'Y + dB'Y')J, \\ S &= AY + mBX + nBX', \qquad m + n = 1. \end{aligned} \tag{44}$$

If $a = b = c = n = 0$, $d = m = 1$, we have the commutative algebra cited in §1, which is a division algebra if c_n is not the square of the norm of a number of $F(J)$.

16. Algebras of order 2 without a modulus.

I. Let e^2 be a scalar multiple of e for every element e. As in the proof of Lemma 1,

$$e_1^2 = ae_1, \qquad e_2^2 = be_2, \qquad e_1e_2 + e_2e_1 = be_1 + ae_2.$$

Hence if $e = xe_1 + ye_2$, $e^2 = (ax + by)e$.

Let a and b be not both zero. By symmetry, take $a \neq 0$. Taking e_1/a as a new e_1, we have $a = 1$. The idempotent elements

$$e = (1 - by)e_1 + ye_2, \qquad E = (1 - bY)e_1 + Ye_2$$

are independent if $Y \neq y$. Let $e_1e_2 = je_1 + ke_2$. The conditions for $eE = 0$ are

$$1 - bY + Yj - yj = 0, \qquad Yk + y - yk = 0.$$

These have unique solutions y, Y with $Y \neq y$ if $j + bk - b \neq 0$. Then

$$e^2 = e, \qquad E^2 = E, \qquad eE = 0, \qquad Ee = e + E. \tag{45}$$

II. In the contrary case, we have (20)–(26) after suppressing scalars and the first equation (23). The condition for $S = J$ is now a homogeneous quadratic in b, c. Instead, we attempt to make $T = K$, given $t \neq k$; the condition is $b = mc$, $m = (s - j)/(t - k)$. Then $D \neq 0$ if $c \neq 0$ and

$$m^3 + m^2(k + t) + m(p - j - s) - n \neq 0. \tag{46}$$

In this case we may therefore take $t = k$.

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